METOPROLOL

Tartrate ≠ succinate

Do not stop suddenly

CHEMICAL NAME (RS)-1-(isopropylamino)-3-[4-(2-methoxyethyl)phenoxy]propan-2-ol (tartrate or succinate)

CLASS Cardioselective β1-adrenoceptor blocker

TRADE NAMES Lopressor (tartrate) · Betaloc · Toprol-XL (succinate) · Seloken

THERAPEUTIC USES / INDICATIONS

- Hypertension; angina.
- Heart failure (succinate ER only).
- Acute MI and secondary prevention.
- Arrhythmias (AF rate control, SVT); migraine prophylaxis; thyrotoxicosis.

ADMINISTRATION

- **PO** tartrate twice daily with food; succinate ER once daily, swallow whole (may split along score).
- **IV** 5 mg bolus, repeated at about 2-min intervals (max 15 mg) with BP, HR and ECG monitoring.

DOSAGES

- **Hypertension:** tartrate 50–100 mg twice daily; succinate ER 25–100 mg once daily initially (max 400 mg/day).
- **Heart failure (succinate):** start 12.5–25 mg daily, titrate to 200 mg.
- **Early MI (IV):** three 5 mg bolus injections about 2 min apart (15 mg); AF rate control per protocol.
- **Post-MI:** 50 mg every 6 h for 48 h, then 100 mg twice daily.

PHARMACOKINETICS

- **Onset:** PO 1 h; IV 5 min. **Duration:** 6–12 h (tartrate); 24 h (succinate).
- Metabolised by CYP2D6 (poor metabolisers ↑ levels).
- Excreted in urine. **t½:** 3–7 h.

PHARMACODYNAMICS

- Selective **β1 blockade** → ↓ HR, contractility, oxygen demand, renin.
- Moderate lipophilicity (some CNS effects).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fatigue, dizziness.
- Bradycardia.
- Cold extremities, sleep disturbance.

ADVERSE EFFECTS (SERIOUS)

- AV block, bradycardia, hypotension.
- Heart failure exacerbation.
- Bronchospasm (high doses).
- Masked hypoglycaemia; rebound on withdrawal.

CONTRAINDICATIONS & CAUTIONS

- Cardiogenic shock; severe bradycardia; 2nd/3rd-degree AV block; sick sinus.
- Decompensated heart failure.
- Severe peripheral arterial disease.
- **Caution:** asthma, diabetes.

KEY INTERACTIONS

- **CYP2D6 inhibitors** (fluoxetine, paroxetine, bupropion): ↑ levels.
- **Verapamil, diltiazem:** heart block.
- **Clonidine:** rebound.
- **Antidiabetics:** masked hypoglycaemia.

NURSING MANAGEMENT

- Check HR and BP before giving; ECG monitoring for IV.
- Confirm correct salt (tartrate vs succinate).
- Daily weight in HF.
- Monitor glucose in diabetics.
- Taper on stopping.

PATIENT EDUCATION

- Swallow ER tablets whole.
- Do not stop suddenly.
- Report slow pulse, fainting, breathlessness.
- Take with food (tartrate).

KEY POINTS

- β1-selective blocker; CYP2D6 metabolism.
- IV for AF rate control.
- Only succinate ER proven in HF.
- Taper to stop.

SOURCES CHECKED **[302]** Metoprolol tartrate 100mg film-coated tablets (UK SmPC) **[303]** Metoprolol Succinate, oral (US label) **[304]** Metoprolol Tartrate, oral (US label) **[305]** Metoprolol Tartrate, oral (US label, Bryant Ranch Prepack) **[306]** Metoprolol, IV (US label)

Full references p. 432–457